

Global Supply Chain Fragility in Complex and Small-World Economic Networks

In our everyday life, supply chains play a crucial role in ensuring access to resources. The global economy relies on vast, interconnected supply chains, which enable us to be more efficient, and make room for specialization, but there are also risks in interconnection, this risk is system vulnerability. In the past few years, we have been experiencing shocks (targeted and untargeted) on a local supply chain, which quickly escalated into a global scale problem, like the Covid-19 pandemic and the Red Sea shipping crisis (IMF blog, 2024). These supply chains show a Small-World like behavior: densely connected (high clustering coefficients), with some long ties (global connections, but short average path), to reduce path lengths (Hearnshaw et al., 2013).

Studies in economic complexity and network science show us that the structure of production networks determines the patterns of growth and vulnerability of a network. Hidalgo and Hausman, 2009 show that countries' inter-industry connections and export baskets determine their ability to react to changing environment. After Covid-19 Guan et al., (2020) found that the complexity of global supply chains magnifies the losses beyond the direct effects of the pandemic. Next to the Small-World behavior, global supply chains show scale-free patterns, when combined, implying high efficiency, but small robustness (Barabási, 2016).

In the literature there are lot of findings about the volume of trade, and the macroeconomic features, I think less is known in the field of networks. The literature doesn't emphasize which network properties (clustering, centrality, modularity) can strengthen the networks resilience, which could make the global supply chain more robust. A possible research question can be: Which network property makes a supply chain more robust in a small-world network? Understanding this question could help determine whether shortening paths, or diversifying connections would be a more efficient way to strengthen supply chains.

Understanding the proposed question could be important for multinational companies, governments and industrial organizations. Applying network properties to better understand the weaknesses of a given supply chain can yield safer and more efficient trade between firms and countries. Firms and countries can map their supply chains and prepare in a way to detect problematic suppliers, which enables them to make targeted resilience strategies, which are more efficient than broad protection. Institutions like the World Bank and WTO could use these findings to monitor global trade, and to ensure stability across the World.

Appendix

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